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## **Set 13: Geographic Data**

### **Question 1: Map Chart**

#### **Aim**

To create a map chart showing the distribution of cities and label its elements.

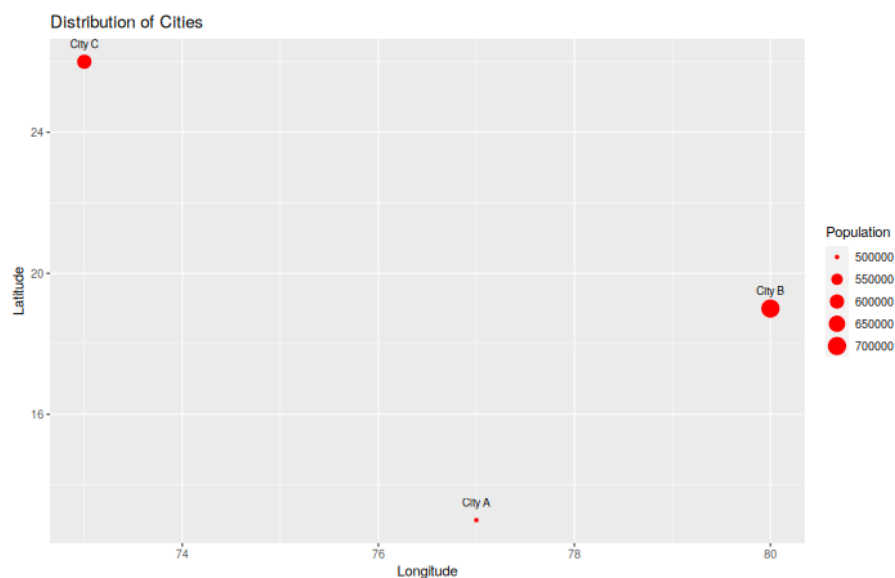
#### **Procedure**

1. Load ggplot2 and create the data frame (with assumed longitude and latitude).
2. Plot cities as points on a longitude-latitude map, sized by population.
3. Add city labels, title and axis labels, then display the map.

#### **Code**

```
library(ggplot2); library(gridExtra)
d <- data.frame(City=c("City A","City B","City C"),
  Population=c(500000,700000,600000), Temp=c(75,68,80),
  Elevation=c(1000,800,1200), Lon=c(77,80,73), Lat=c(13,19,26)) # lon/lat
assumed
m <- ggplot(d, aes(Lon, Lat, size=Population, label=City)) +
  geom_point(color="red") + geom_text(vjust=-1.5, size=3) +
  labs(title="Distribution of Cities", x="Longitude", y="Latitude")
print(m)
```

#### **Output**



*Fig 1: Map chart of cities*

#### **Result**

The map chart was created. Each city is a labelled point; larger points mean larger population (City B is the biggest).

## Question 2: Scatter Plot

### Aim

To explore the relationship between average temperature and population size using a scatter plot.

### Procedure

1. Plot Temperature on the x-axis and Population on the y-axis.
2. Add city labels and a trend line.
3. Find the correlation with `cor()` and interpret it.

### Code

```
s <- ggplot(d, aes(Temp, Population)) + geom_point(size=4, color="blue") +  
  geom_smooth(method="lm", se=FALSE) + geom_text(aes(label=City), vjust=-1)  
+  
  labs(title="Temperature vs Population", x="Avg. Temperature (F)",  
y="Population")  
print(s)  
cat("Correlation:", round(cor(d$Temp, d$Population), 2))
```

### Output

#### Console output:

Correlation: -0.58

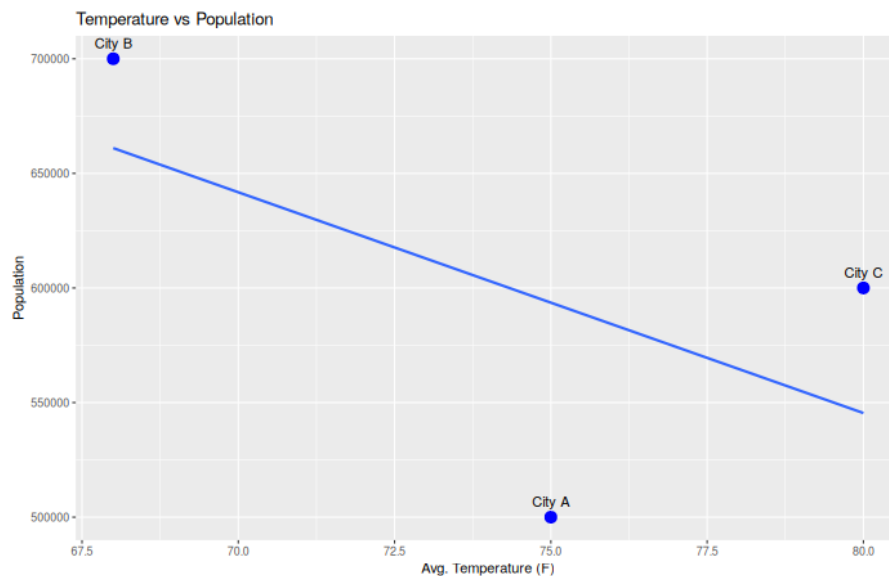


Fig 2: Scatter plot – Temperature vs Population

### Result

The scatter plot was created. Insight: the correlation is -0.58 (negative), so the hotter cities (C: 80°F, A: 75°F) have smaller populations than the coolest city B (68°F, 700000). With only three cities this is a weak indication.

### Question 3: Table

#### Aim

To build a labelled table displaying the geographic data of each city.

#### Procedure

1. Select the City, Population, Temp and Elevation columns.
2. Print the data frame and draw it as a table using tableGrob().

#### Code

```
print(d[, 1:4])  
t <- tableGrob(d[, 1:4], rows=NULL)  
grid.arrange(t)
```

#### Output

##### Console output:

```
City Population Temp Elevation  
1 City A      500000 75      1000  
2 City B      700000 68        800  
3 City C      600000 80      1200
```

| City   | Population | Temp | Elevation |
|--------|------------|------|-----------|
| City A | 500000     | 75   | 1000      |
| City B | 700000     | 68   | 800       |
| City C | 600000     | 80   | 1200      |

*Fig 3: Table of geographic data*

#### Result

The table displaying city, population, average temperature and elevation was created with labelled column headings.

## Question 4: Dashboard

### Aim

To combine the map chart, scatter plot and table into one dashboard for exploring the geographic data.

### Procedure

1. Use the map (m), scatter plot (s) and table (t) from the earlier questions.
2. Arrange all three in one panel using `grid.arrange()`.

### Code

```
grid.arrange(m, s, t, ncol=2)
```

### Output

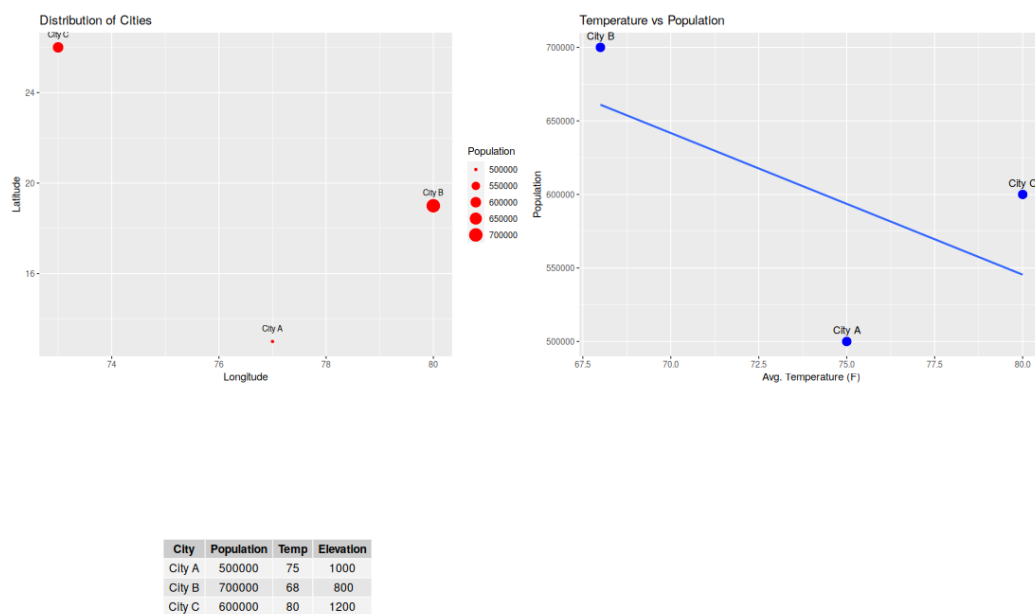


Fig 4: Dashboard (map + scatter plot + table)

### Result

The dashboard combining the map chart, scatter plot and table was created in R.